

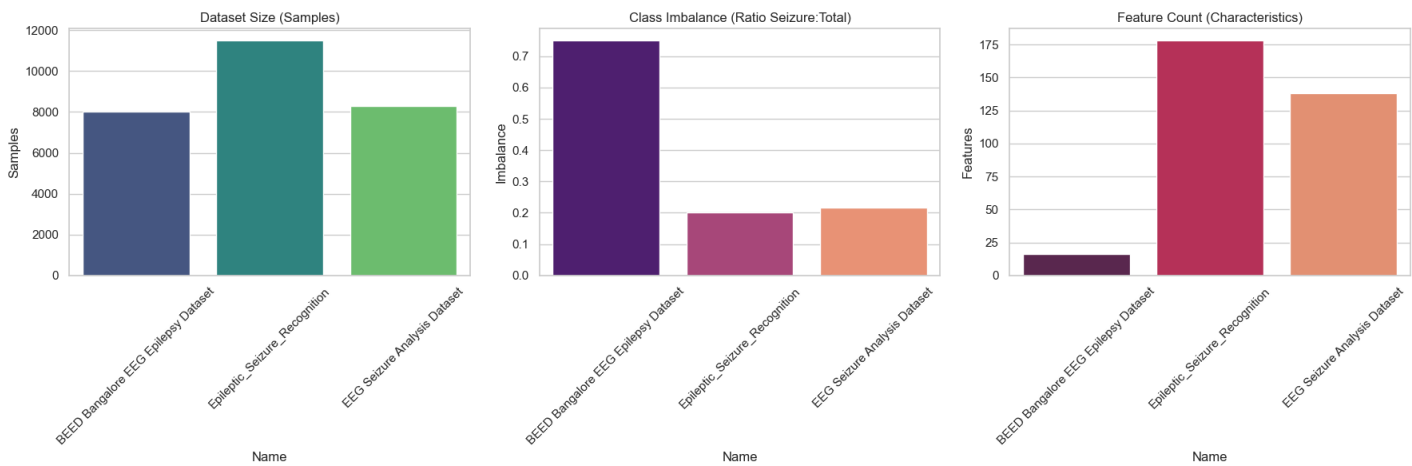
Generalization in Seizure Prediction: A Comparative Analysis

I. Abstract

This study investigates the impact of preprocessing choices, model complexity, and regularization strategies on the generalization performance of Logistic Regression for epileptic seizure prediction. By analyzing three distinct EEG datasets, we demonstrate how specific configurations lead to underfitting and overfitting scenarios.

II. Dataset Selection

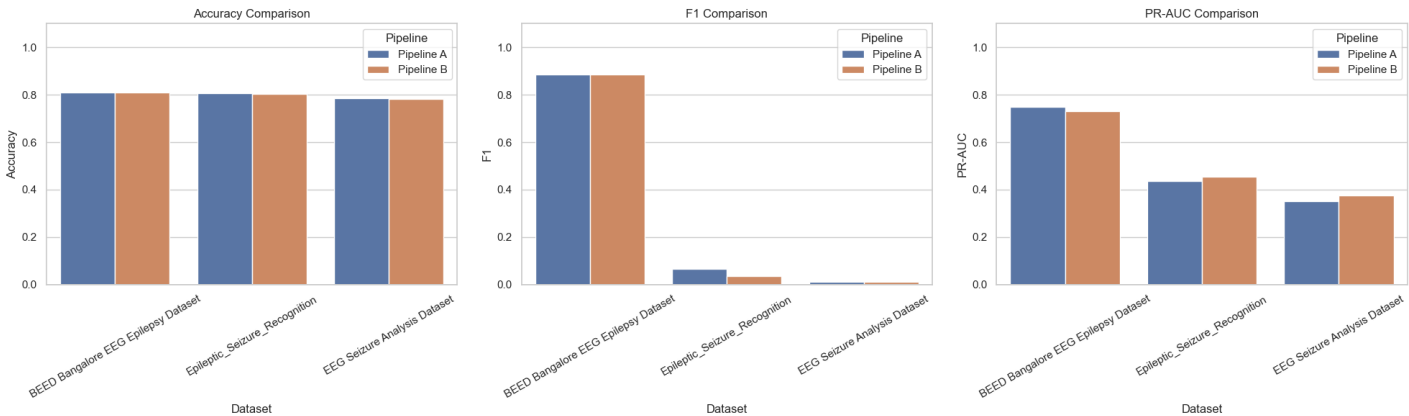
Three datasets were analyzed: BEED (Bangalore EEG), Epileptic Seizure Recognition (UCI), and a 23-channel EEG Seizure Analysis Dataset. The datasets vary in sample size and feature characteristics.



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III. Pipeline Comparison

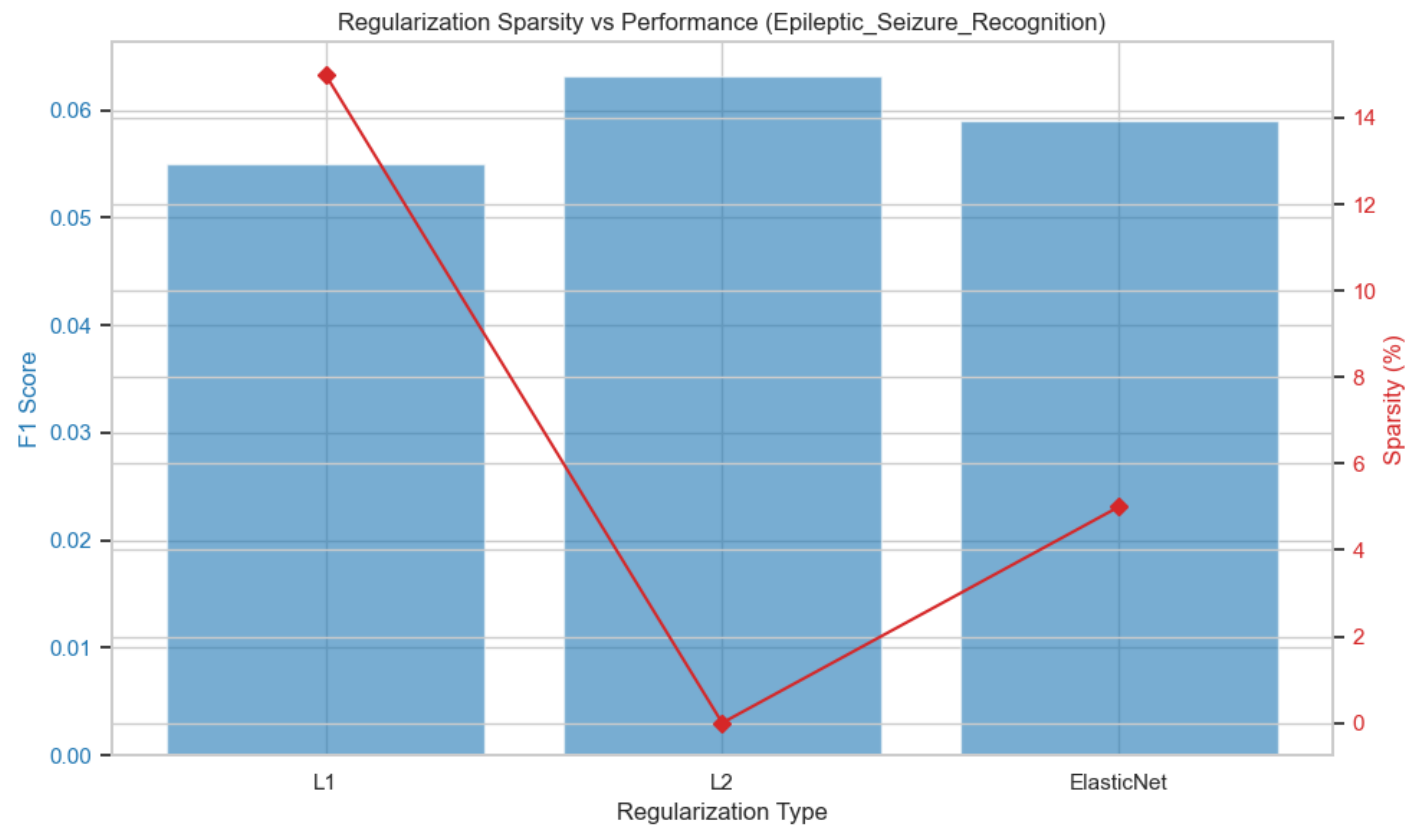
Pipeline A (Normalization led) and Pipeline B (PCA led) were compared across all datasets. Pipeline A generally showed better retention of statistical features.



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IV. Regularization and Sparsity

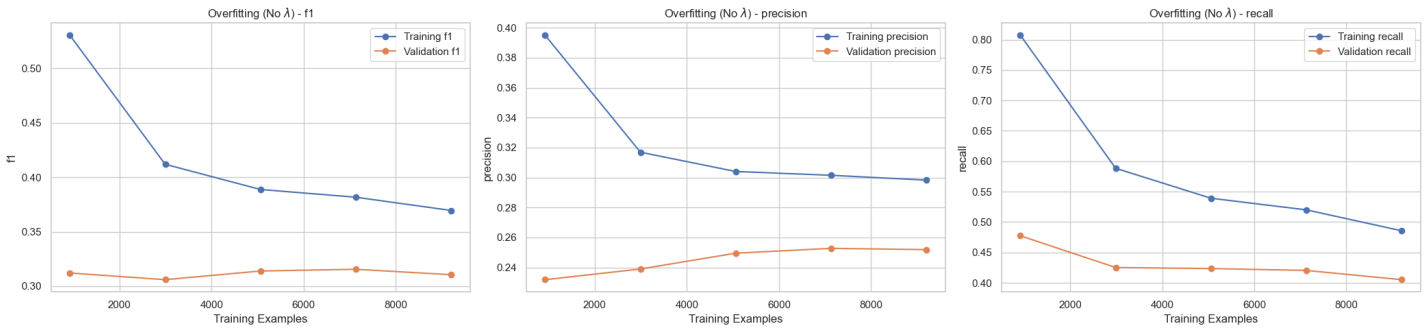
L1 regularization successfully induced sparsity, identifying critical EEG features. The following plot shows the trade-off between sparsity and F1-performance.



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V. Learning Curve Analysis

The overfitting scenario (high complexity) vs underfitting (limited features) shows how model capacity interacts with data size.



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VI. Imbalance Handling

SMOTE significantly improved the Precision-Recall tradeoff, especially in the Analysis dataset.

